Layer 8: Probe Architecture and Conformational Dynamics

WLC Tether Statistics, Conformationally-Averaged Electron Transfer, and the Steric Gate on MMP-9 Access

Eton iGEM 2026, Dry Lab | Layer 8 of 17-Layer Biosensor Pipeline | `probe_conformation.py` · `average_ket.py` · `steric_gate.py`

Layer 8 is the physical-conformation layer sitting between the SAM chemistry that sets probe surface density (Layer 7) and the two layers that consume a probe geometry as an input: Layer 9 (surface enzyme kinetics, which needs an effective probe-layer thickness δ_{eff} to convert a solution-phase K_m into surface units) and Layer 10 (Marcus-theory electron transfer, which needs a tunnelling distance d between the methylene-blue (MB) reporter and the gold surface). Because $k_{ET} \propto e^{-\beta d}$, the electron-transfer rate is exponentially sensitive to the conformational fluctuations of the flexible tether; the central physical point of this layer is that the population-averaged rate $\langle k_{ET} \rangle$ is *not* well approximated by evaluating k_{ET} at the mean tether distance $\langle d \rangle$, a placeholder both Layer 9 and Layer 10 flag as still outstanding (Layer 9 § IV, Eq. IV.2; Layer 10 § IX, Eq. IX.1).

Architecture note. The original planning notes (`contents.md`, Layer 8) specified two candidate probes – an HNE-channel probe (AAPV↓SGIG) and an MMP-9-channel probe (IPRT↓LTAA) – plus a scrambled-sequence negative control, on a PEG4-or-PEG6 spacer. Layers 7, 9, 10 and 13 have since fixed the fabricated design: a *single* MMP-9 channel carrying the peptide SGKIPRTLTA on a PEG6 spacer (Layer 7 § I), with a D-enantiomer stereospecific negative control rather than a sequence scramble (Layer 13 § III). There is no separate HNE probe; HNE is treated purely as a possible interferent on the single MMP-9 channel (Layer 9 § X, Layer 15 § IV). This report therefore supersedes the two-probe/PEG4-or-6 framing in the planning notes and applies the requested WLC/Monte-Carlo/steric-gate physics to the fabricated architecture: **Ac-Cys-PEG6-SGKIPRTLTA-PEG2-Lys(MB)-NH$_2$**. The underlying polymer-physics model (worm-like-chain tether, Monte Carlo $P(r)$, brush/mushroom steric gating) is sequence-agnostic; only the contour-length inventory and the specific charged residues in § V below are updated relative to the planning notes.

No paper is yet present in `papers/layer_08/`; nearly every numeric input below is therefore a literature estimate carried over from the planning notes or drawn from a value already confirmed in an upstream layer, and is flagged accordingly.

I Fabricated Probe Architecture and Contour-Length Inventory

Segment inventory

The Au–MB tether is modelled as five worm-like-chain (WLC) segments in series: the Cys anchor residue, the PEG6 spacer (Layer 7 § I; supersedes the PEG4-or-6 choice in the planning notes), the SGKIPRTLTA substrate (10 residues, fabricated sequence, Layer 7 § I; supersedes AAPVSGIG/IPRTLTAA), the PEG2 linker, and the Lys(MB) label linker. Segment contour lengths use the planning-notes monomer values – PEG monomer $b_{\text{PEG}} = 0.36$ nm (extended C–C–O bond length) and extended β-strand $b_{\text{aa}} = 0.35$ nm/residue – neither of which is sourced from a paper in `papers/layer_08/`; both are standard polymer-physics textbook values and are used here as in the planning notes.

Segment	L (nm)	l_p (nm)	Basis
Cys anchor	0.50	0.60	Peptide-type (planning-notes estimate)
PEG6 spacer	2.16	0.38	Oesterhelt et al. 1999 [1] (AFM force spectroscopy)
SGKIPRTLTA (10 aa)	3.50	0.60	Unstructured-peptide central estimate, range 0.4–1.0 nm
PEG2 linker	0.72	0.38	Oesterhelt et al. 1999 [1]
Lys(MB) linker	0.50	0.60	Peptide-type (planning-notes estimate)
Total	**7.38**	–	L_c = full extended contour length

Table 1: Contour-length inventory for the fabricated tether Ac-Cys-PEG6-SGKIPRTLTA-PEG2-Lys(MB)-NH$_2$ (`probe_conformation.py`, Sec. I). L_c is the fully extended length; the mean end-to-end distance is much smaller (§ III).

SGKIPRTLTA carries one proline (P5) that may add local backbone stiffness, but no confirmed β-sheet-forming motif; the peptide persistence length is therefore held at the same central literature estimate ($l_p \approx 0.6$ nm, range 0.4–1.0 nm) used for the unstructured Cys and Lys(MB) linker segments, pending a sequence-specific measurement (Limitation 1).

II Worm-Like-Chain Model of the Heterogeneous Tether

Each segment's own mean-squared end-to-end distance follows the Kratky-Porod WLC relation,

$$\langle r_i^2 \rangle = 2\, l_{p,i}\, L_i \left(1 - \frac{l_{p,i}}{L_i}\left(1 - e^{-L_i/l_{p,i}}\right)\right). \tag{II.1}$$

The five segments are chemically distinct (PEG polyether vs. peptide backbone) and are treated as orientationally decorrelated at each junction, i.e. the total mean-squared end-to-end distance is the sum of the individual segment values, $\langle r^2 \rangle_{\text{tot}} = \sum_i \langle r_i^2 \rangle$. An effective single persistence length $l_{p,\text{eff}}$ for the whole concatenated chain is then defined by variance-matching against a homogeneous WLC of length L_c:

$$\langle r^2 \rangle (L_c,\, l_{p,\text{eff}}) \equiv \langle r^2 \rangle_{\text{tot}}. \tag{II.2}$$

Solving Eq. (II.2) numerically (`probe_conformation.py`) gives $l_{p,\text{eff}} = 0.40$ nm and $\sqrt{\langle r^2 \rangle_{\text{tot}}} = 2.35$ nm. The ratio $L_c / l_{p,\text{eff}} \approx 18.7$ places the tether firmly in the short-chain, non-Gaussian regime flagged in the planning notes (§ 8.3): the Marko–Siggia/Gaussian-chain closed forms are not expected to be quantitatively accurate, which motivates the direct Monte Carlo sampling in § III.

III Monte Carlo Distance Ensemble $P(r)$

Each segment is discretised into steps of length $l_0 = 0.05$ nm ($\ll l_{p,i}$ for every segment) using the standard bead-tangent algorithm: successive tangent vectors are related by a random rotation whose polar angle is drawn from the von-Mises-Fisher-type density $p(\cos\theta) \propto \exp[(l_{p,i}/l_0)\cos\theta]$, reproducing the Kratky-Porod tangent-tangent correlation $\langle \cos\theta \rangle \approx e^{-l_0/l_{p,i}}$ for $l_0 \ll l_{p,i}$. The five segments are concatenated into a single 3-D path from the Au anchor to the MB label.

Surface exclusion. The Cys-thiol anchor is chemically bonded to the gold at $z = 0$; the electrode is an impenetrable half-space ($z < 0$ is metal). Any Monte Carlo path that dips below $z = 0$ therefore corresponds to the chain passing through the electrode and is unphysical. Each realisation is rejected and resampled until the full path satisfies $z \geq 0$ throughout (reflecting-boundary WLC; this boundary condition is standard for surface-tethered redox probes, cf. Rant et al. 2004 [3] for the analogous DNA case). Without this constraint the unconstrained ensemble mean is $\langle r \rangle = 2.42$ nm; imposing the physical boundary condition shifts it upward to $\langle r \rangle = 2.70$ nm, since it removes a subset of the shortest-r excursions.

Quantity (12,000 realisations)	Value
$\langle r \rangle$	2.70 nm
$\sqrt{\langle r^2 \rangle}$	2.85 nm
Median r	2.69 nm
10th / 90th percentile	1.47 / 3.92 nm

Table 2: Monte Carlo $P(r)$ summary statistics, surface-excluded ensemble (`probe_conformation.py`).

IV Conformationally-Averaged Electron-Transfer Rate

The tunnelling distance entering Layer 10's Marcus/Smalley rate law is the full 3-D MB–Au distance r, not its surface-normal projection (planning notes § 8.3). Using Layer 10's own parameters unmodified (Eq. II.5 of the Layer 10 report, Smalley et al. 1995 [confirmed via Layer 10]),

$$k_{ET}(r) = k_0\, e^{-\beta r}, \qquad k_0 = 6.0 \times 10^8 \text{ s}^{-1}, \quad \beta = 0.95 \text{ Å}^{-1} = 9.5 \text{ nm}^{-1}, \tag{IV.1}$$

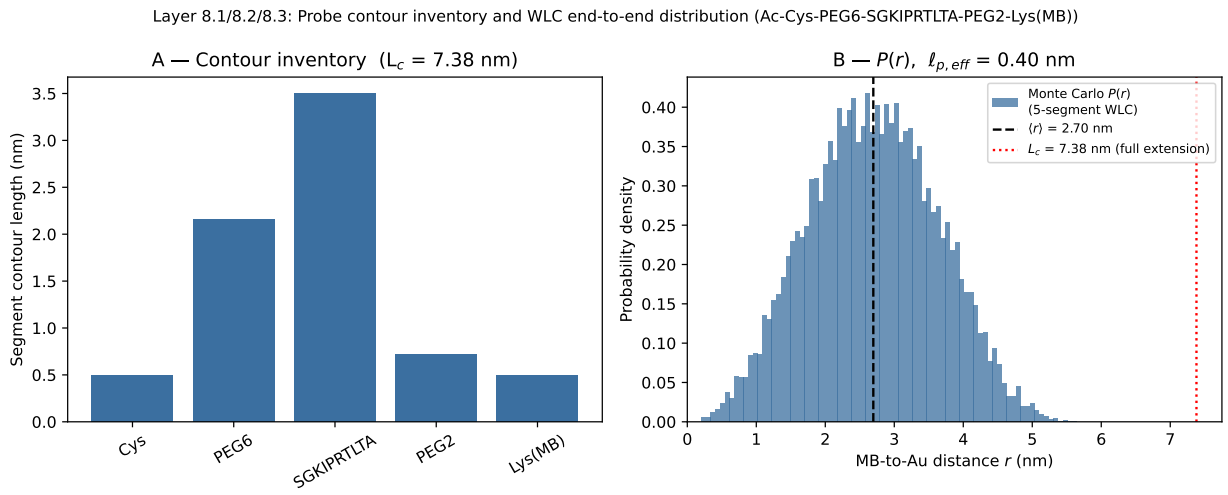


Figure 1: Left: contour-length inventory by segment. Right: Monte Carlo $P(r)$ (surface-excluded, 12,000 realisations) against the fully extended contour length L_c.

the population-averaged rate is

$$\langle k_{ET} \rangle = k_0 \int_0^{L_c} e^{-\beta r}\, P(r)\, \mathrm{d}r \approx k_0\, \overline{e^{-\beta r}}\Big|_{\mathrm{MC}}. \tag{IV.2}$$

Because $k_{ET}(r)$ is a convex (exponential) function of r, Jensen's inequality guarantees $\langle k_{ET} \rangle > k_{ET}(\langle r \rangle)$; the planning notes anticipate this directly (§ 8.3: "the mean $\langle k_{ET} \rangle$ is not simply k_{ET} evaluated at the mean distance $\langle d \rangle$"). The Monte Carlo ensemble from § III gives:

Quantity	Value
$k_{ET}(\langle r \rangle)$ [naive point estimate]	4.5×10^{-3} s^{-1}
$\langle k_{ET} \rangle$ [Monte Carlo average over $P(r)$]	9.0×10^4 s^{-1} (95% CI $[6.3, 12.2] \times 10^4$)
$\langle k_{ET} \rangle / k_{ET}(\langle r \rangle)$	$\sim 2 \times 10^7$

Table 3: Conformationally-averaged vs. naive point-estimate k_{ET} (`average_ket.py`). The 95% CI is a 1000-resample bootstrap over the Monte Carlo ensemble.

The naive point estimate is not merely biased low; at $\langle r \rangle = 2.70$ nm it sits deep in the exponential tail of $k_{ET}(r)$, where the scale over which k_{ET} changes by an order of magnitude is only $\ln(10)/\beta \approx 0.24$ nm – an order of magnitude narrower than the width of $P(r)$ itself ($\sim$ 1.4 nm, 10th–90th percentile). The population average is consequently dominated not by the median configuration but by the shortest-r $\sim$5–10% tail of the ensemble. A semi-analytic Gaussian-chain cross-check using the same $l_{p,\mathrm{eff}}$ (no surface exclusion) gives $\langle k_{ET} \rangle_{\mathrm{analytic}} = 4.3 \times 10^5$ s^{-1}, about 4.8× higher than the surface-excluded Monte Carlo value – consistent with the analytic form lacking the physical reflecting boundary and therefore over-weighting the unphysical near-surface tail (Limitation 3).

Feed to Layer 10. Layer 10 § IX (Eq. IX.1) currently evaluates $k_{ET}^0(d)$ at a single placeholder $\langle d \rangle$ pending this layer. The value that should replace that placeholder is $\langle k_{ET} \rangle = 9.0 \times 10^4$ s^{-1} (before the temperature correction $f_T = 1.38$ and Frumkin correction of Layer 10 § VII–VIII), which places the standard-state device in Layer 10's "quasi-reversible" SWV regime (Layer 10 Table 3) – between their tabulated $d = 1.0$ and $d = 1.5$ nm rows, despite $\langle r \rangle$ itself being 2.70 nm. This is the clearest illustration in this layer of why a distributional treatment, not a point estimate at $\langle d \rangle$, is required to select the SWV operating parameters in Layer 11.

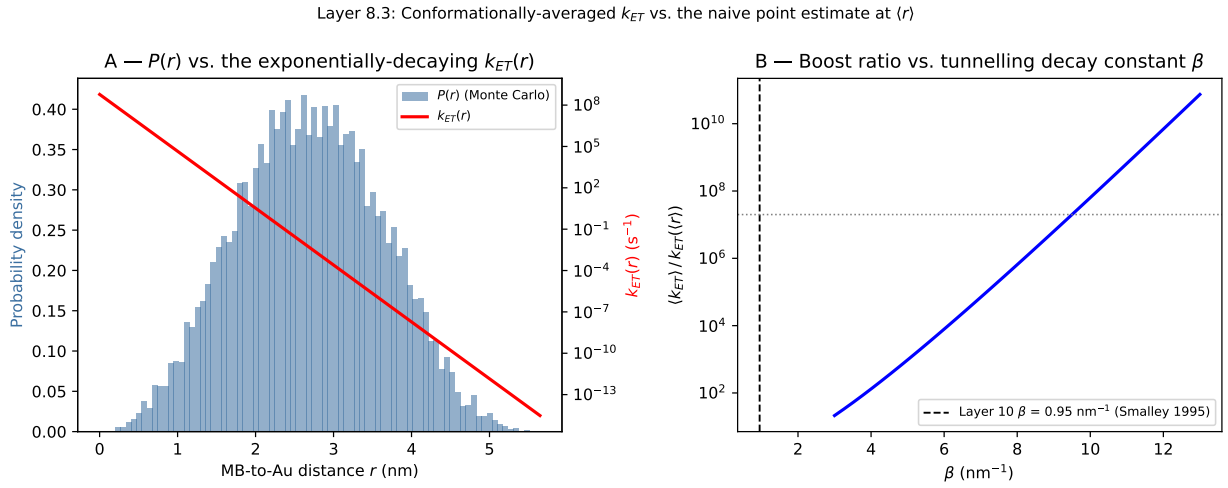


Figure 2: Left: $P(r)$ against the exponentially-decaying $k_{ET}(r)$ on a log axis. Right: sensitivity of the $\langle k_{ET} \rangle / k_{ET}(\langle r \rangle)$ boost ratio to the tunnelling decay constant β; the qualitative conclusion (large boost) holds across the plausible β range, not only at Layer 10's specific value.

V Electrostatic Bending from the Cationic SGKIPRTLTA Tether

The planning notes' IPRTLTAA-specific electrostatic-gating argument (§ 8.7) was written for a probe carrying a single cationic residue (Arg, P2). The fabricated SGKIPRTLTA sequence in fact carries *two* residues that are cationic at pH 7.4: Lys (K, position 3) and Arg (R, position 6), so this correction applies with, if anything, greater force to the real device than to either originally-planned sequence.

During the cathodic MB-reduction sweep the electrode is negatively polarised (Layer 10 § VI: $E^{\circ\prime}_{MB} \approx -0.29$ V vs. Ag/AgCl at pH 7); the two cationic sidechains are then electrostatically attracted toward the surface, which is expected to bend the tether and contract the effective MB–Au bridge length by some Δd, increasing k_{ET} by $\exp(\beta \Delta d)$ (same functional form used for the analogous Ca^{2+}/PEG argument in Layer 7 § IV). At physiological ionic strength the Debye screening length is $\lambda_D \approx 0.8$ nm (Layer 0), which sets the natural length scale over which such electrostatic contraction could act:

Assumed contraction Δd	Fold change in k_{ET}, $e^{\beta \Delta d}$
0.5 nm	$1.2 \times 10^2\times$
0.8 nm ($\approx \lambda_D$)	$2.0 \times 10^3\times$
1.0 nm	$1.3 \times 10^4\times$

Table 4: Order-of-magnitude sensitivity of k_{ET} to a hypothetical electrostatically-driven tether contraction, using Layer 10's β. No Δd value is confirmed for this device (Limitation 5).

Given the exponential sensitivity already established in § IV, even a sub-Debye-length contraction could be electrochemically significant. Unlike the Ca^{2+} artefact (Layer 7 § IV), this effect does *not* automatically cancel in the ratiometric signal chain, because it depends on the electrode potential during the scan (present identically on active and D-control electrodes, which are exposed to the same scan waveform, but not present at all outside the cathodic window) – it is a potential-dependent, not a probe-density-dependent, artefact, and should enter Layer 10 as an explicit $k_{ET}(E)$ correction term rather than being assumed to cancel. No Δd value is confirmed from any paper in `papers/layer_08/`; a full Odijk–Skolnick–Fixman polyelectrolyte-persistence-length treatment (cf. Netz & Joanny 1999 [4]) is left as future work (Limitation 5).

VI SAM Density and the Brush/Mushroom Transition

At high probe surface density, neighbouring tethers sterically repel one another and extend away from the surface (brush regime); at low density, tethers behave as isolated coils (mushroom regime). The planning-notes crossover density is

$$\Gamma^* = \frac{1}{4\pi l_{p,\mathrm{eff}}^2}, \tag{VI.1}$$

which, using this layer's $l_{p,\mathrm{eff}} = 0.40$ nm, gives $\Gamma^* \approx 85$ pmol/cm^2 – roughly 17–40× above Layer 7's device-planned target window ($\Gamma_p = 2$–5 pmol/cm^2) and well above even the close-packed alkanethiol limit fraction relevant here. **The tether is therefore in the mushroom regime throughout the entire steric-window-compatible density range**; the Daoud–Cotton brush scaling in the planning notes (§ 8.4) is not the operative regime for this device and $d_{\mathrm{eff}}(\Gamma_p)$ is correspondingly nearly flat across the target window:

$$d_{\mathrm{eff}}(\Gamma_p) = r_g \left(1 + \frac{\Gamma_p}{\Gamma^*}\right)^{1/3}, \qquad r_g \equiv \sqrt{\langle r^2 \rangle_{\mathrm{tot}}/6} = 0.96 \text{ nm}, \tag{VI.2}$$

giving $d_{\mathrm{eff}} = 0.97$ nm at $\Gamma_p = 3$ pmol/cm^2 against $r_g = 0.96$ nm at $\Gamma_p \to 0$ – a $< 1.5\%$ brush-crowding correction across the entire target window (Fig. 3C). This is a useful negative result: SAM-density-driven brush stretching is *not* a first-order effect on k_{ET} for this device, in contrast to the steric blocking of enzyme access treated next, which *is* a first-order effect over the same density range.

VII Steric Gate: Enzyme Accessibility vs. Probe Density

Cleavage-site height h_{cut}

The exact cleavage bond within SGKIPRTLTA is not established in any paper in `papers/layer_07/` or `papers/layer_09/` (Kridel 2001 [confirmed via Layer 9] reports bulk k_{cat}/K_m for the intact decapeptide substrate, not the cleavage position). h_{cut}, the height of the scissile bond above the gold, is therefore bounded rather than pinned:

$$h_{\mathrm{cut}} \in \left[\, L_{\mathrm{Cys}} + L_{\mathrm{PEG6}},\;\; L_{\mathrm{Cys}} + L_{\mathrm{PEG6}} + L_{\mathrm{peptide}} \,\right] = [\,2.66,\; 6.16\,] \text{ nm}. \tag{VII.1}$$

This range should be narrowed once the P1–P1′ bond is identified experimentally or from a MEROPS-style specificity search (Limitation 4).

Enzyme footprint and $\Gamma_{\mathrm{steric}}^{\mathrm{max}}$

Layer 7 (§ "Probe spacing constraint from enzyme size") already estimates the MMP-9 hydrodynamic radius as $r_{\mathrm{MMP9}} \approx 3$–$4$ nm from its molecular weight (∼92 kDa) under a globular-protein approximation – itself not confirmed from any paper in `papers/layer_07/` – and derives a nearest-neighbour spacing criterion $\ell \gtrsim 2\, r_{\mathrm{MMP9}}$. This layer applies the planning notes' independent area-exclusion criterion, $A_{\mathrm{probe}} = 1/\Gamma_p > \pi r_{\mathrm{enzyme}}^2$, over the same r_{MMP9} range, as a cross-check against Layer 7's own number:

r_{MMP9}	$\Gamma_{\mathrm{steric}}^{\mathrm{max}}$, area-exclusion (this layer)	$\Gamma_{\mathrm{steric}}^{\mathrm{max}}$, spacing (Layer 7)
3.0 nm	5.87 pmol/cm^2	4.61 pmol/cm^2
3.5 nm	4.31 pmol/cm^2	3.39 pmol/cm^2
4.0 nm	3.30 pmol/cm^2	2.59 pmol/cm^2

Table 5: Two independent geometric estimates of $\Gamma_{\mathrm{steric}}^{\mathrm{max}}$, agreeing to within ∼25–30% across the plausible r_{MMP9} range (`steric_gate.py`).

The two independent geometric arguments – area exclusion (this layer) and nearest-neighbour spacing (Layer 7) – agree to within ∼25–30% at every r_{MMP9} value, which is reassuring given they use unrelated packing assumptions (circular exclusion zone vs. square-lattice spacing).

A genuine finding: the target window collides with the steric ceiling

Layer 7 sets a device-planned target of $\Gamma_p = 2$–5 pmol/cm^2 without reference to this layer's steric calculation (it predates it). Combining the two:

$$f_{\text{steric}}(\Gamma_p) = \max\left(0,\ 1 - \frac{\Gamma_p}{\Gamma^{\max}_{\text{steric}}}\right). \tag{VII.2}$$

Γ_p (pmol/cm^2)	f_{steric}, $r = 3.0$ nm	$r = 3.5$ nm	$r = 4.0$ nm
2	0.66	0.54	0.40
3	0.49	0.31	0.09
5	0.15	0.00	0.00

Table 6: Steric accessibility factor across Layer 7's target window (`steric_gate.py`).

At the mid-to-upper end of Layer 7's own target range, MMP-9 access is already substantially degraded ($f_{\text{steric}} = 0.31$ at $\Gamma_p = 3$ pmol/cm^2, $r = 3.5$ nm) and reaches or approaches full steric blockade at $\Gamma_p = 5$ pmol/cm^2 for the mid-to-high end of the plausible enzyme-radius range. This is not a modelling artefact: it is a direct consequence of combining Layer 7's own probe-density target with Layer 7's own enzyme-radius estimate through the same geometric logic Layer 7 used to derive that target in the first place. **Layer 7's target window should be revisited toward its lower end (2–3 pmol/cm^2), or the enzyme-radius estimate underlying both calculations should be confirmed experimentally**, before the upper half of the stated target range is relied upon.

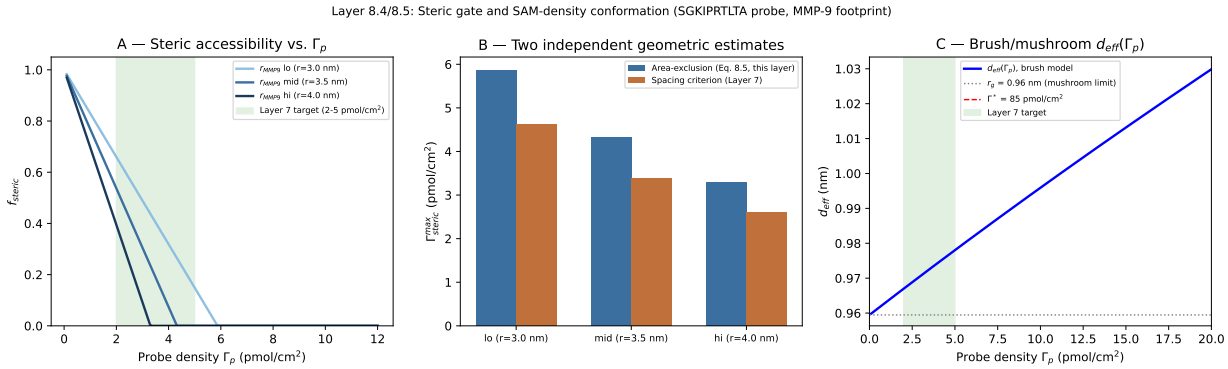


Figure 3: A: steric accessibility f_{steric} vs. probe density across the plausible MMP-9 radius range, with Layer 7's target window overlaid. B: cross-check between this layer's area-exclusion estimate and Layer 7's spacing-based estimate of $\Gamma^{\max}_{\text{steric}}$. C: brush-model $d_{\text{eff}}(\Gamma_p)$, confirming the mushroom regime holds throughout the target window (§ VI).

VIII Feedback to Layer 9: δ_{eff} and the Probe-Saturation Ratio ρ

Layer 9 (§ IV, Eq. IV.2) converts the solution-phase $K_m^{\text{sol}} = 1.28$ mM (Kridel 2001 [confirmed via Layer 9]) into a surface Michaelis constant via

$$K^{\Gamma}_{m,\text{surf}} = K_m^{\text{sol}} \times 10^{-3} \times \delta_{\text{eff}}, \tag{VIII.1}$$

and explicitly flags δ_{eff} as unconfirmed pending this layer (Layer 9 Limitation 5), using a placeholder $\delta_{\text{eff}} = 5$ nm. This layer's mushroom-regime radius-of-gyration estimate at Layer 7's target midpoint ($\Gamma_p = 3$ pmol/cm^2) gives $d_{\text{eff}} \approx 0.97$ nm (§ VI) – a physically motivated candidate for δ_{eff}, roughly 5× smaller than Layer 9's placeholder. Propagating this through Layer 9's own equations:

A five-fold increase in ρ pushes the device further from Layer 9's "intermediate" regime toward its zeroth-order/Langmuir-like limit ($\rho \gg 1$, Layer 9 § IV), where the progress curve $F_k(T)$ becomes linear in time and saturates at late times rather than following a monoexponential. **This is a candidate correction for Layer 9's Eq. IV.3 and Table 1 and should be back-ported there**, though it should be treated as a first-pass

	Layer 9 placeholder ($\delta_{\text{eff}} = 5$ nm)	This layer ($\delta_{\text{eff}} \approx 0.97$ nm)
$K^{\Gamma}_{m,\text{surf}}$	6.4×10^{-13} mol cm^{-2}	1.2×10^{-13} mol cm^{-2}
$\rho = \Gamma_0 / K^{\Gamma}_{m,\text{surf}}$ (at $\Gamma_0 = 2.0 \times 10^{-12}$ mol cm^{-2})	3.1	16.1

Table 7: Effect of this layer's δ_{eff} estimate on Layer 9's probe-saturation ratio ρ (Layer 9 Eq. IV.4, Table 1).

geometric estimate, not a substitute for fitting ρ directly from a measured progress curve once the fabricated electrode exists (Limitation 6) – the same caveat Layer 9 itself already attaches to ρ.

IX Post-Cleavage State

After MMP-9 cleaves SGKIPRTLTA at its (unconfirmed) scissile bond, the surface retains an Ac-Cys-PEG6-(partial peptide) stub carrying no MB, contributing nothing to the SWV peak; the MB-bearing C-terminal fragment is released into solution (negligible bulk contribution, Layer 3 § 3.6). The stub is uniformly shorter than an intact probe, so it is not expected to sterically block enzyme access to remaining intact probes; if anything it marginally increases local free volume. Following the planning notes (§ 8.6), this is treated as a second-order effect, negligible for $F_k < 0.9$, and is not modelled further here.

X Key Outputs

Output	Value (this report)	Feeds
$P(r)$: MB-to-Au distance distribution	Monte Carlo WLC, surface-excluded, $\langle r \rangle = 2.70$ nm, $L_c = 7.38$ nm	Layer 10
$\langle k_{ET} \rangle$	9.0×10^4 s^{-1} (vs. naive $k_{ET}(\langle r \rangle) = 4.5 \times 10^{-3}$ s^{-1})	Layer 10 § IX
h_{cut}	2.7–6.2 nm from Au (scissile bond position unconfirmed)	Layer 9: enzyme access geometry
$\Gamma^{\text{max}}_{\text{steric}}$ (MMP-9)	3.3–5.9 pmol/cm^2 ($r_{\text{MMP9}} = 3$–4 nm)	Layer 9: $K_{m,\text{surf}}$ correction
$f_{\text{steric}}(\Gamma_p = 3)$	0.09–0.49	Layer 9
δ_{eff} (revised)	$\approx$ 0.97 nm, mushroom-regime estimate (Layer 9 placeholder: 5 nm)	Layer 9 Eq. IV.2 – recommend update
$d_{\text{eff}}(\Gamma_p)$, brush model	Mushroom regime throughout target window; $\Gamma^* \approx 85$ pmol/cm$^2 \gg$ target	Layer 10
Electrostatic correction (K3+R6)	Order-of-magnitude only, 10^2–$10^4\times$ over plausible Δd; unconfirmed	Layer 10, potential-dependent term

XI Limitations

1. **Persistence lengths are literature estimates, not measured on the fabricated tether.** $l_{p,\text{PEG}}$ is taken from Oesterhelt et al. 1999 [1] for PEG in water; $l_{p,\text{peptide}}$ is a generic unstructured-peptide central estimate (range 0.4–1.0 nm). No paper in `papers/layer_08/` yet confirms either value, or a sequence-specific stiffness contribution from the SGKIPRTLTA proline.

2. **The additive-variance segment-concatenation approximation is not validated against the exact heterogeneous transfer matrix.** Treating the PEG/peptide junctions as fully orientation-decorrelating is a simplification; a full multi-segment Kratky-Porod transfer-matrix calculation (Marko & Siggia 1995 [2]) would refine $P(r)$ but was judged unnecessary for the order-of-magnitude conclusions drawn here.

3. **The Monte Carlo chain has no excluded volume beyond the single gold-surface half-space constraint.** Peptide backbone self-avoidance and side-chain sterics are not modelled; this could modestly reduce the very-short-r population relative to the values used in § IV, which is the direction the surface-exclusion correction already moved the estimate.

4. **The scissile bond position within SGKIPRTLTA is unconfirmed**, so h_{cut} is reported as a bound (2.7–6.2 nm) rather than a point value; this should be narrowed from a MEROPS-style specificity search or direct MS mapping of the cleavage products.
5. **The electrostatic bending correction (§ V) is order-of-magnitude only.** No Netz–Joanny [4] polyelectrolyte-persistence-length calculation was performed; Δd is not confirmed from any paper in `papers/layer_08/`, and the sign/magnitude of the effect at the fabricated SAM's actual ionic strength and pH is not established.
6. r_{MMP9} **is inherited, unconfirmed, from Layer 7**'s MW-based globular approximation; $\Gamma_{\text{steric}}^{\max}$ and every downstream number in § VII–VIII inherits this uncertainty. A dynamic light scattering or crystal-structure-based estimate of the MMP-9 hydrodynamic radius would materially tighten this layer's central finding (§ VII).
7. **The $\delta_{\text{eff}} \approx 0.97$ nm estimate fed to Layer 9 (§ VIII) is a first-pass geometric value**, not a fit to a measured progress curve; it should be treated as a candidate replacement for Layer 9's placeholder, to be superseded once real $F_k(T)$ data exist.
8. **The β used throughout § IV is Layer 10's alkanethiol-calibrated Smalley 1995 value**, which Layer 10 itself flags (their Limitation 2) as unconfirmed for a PEG/peptide bridge; this uncertainty propagates directly and multiplicatively into every $\langle k_{ET} \rangle$ figure in this report.

References

Papers referenced here are either newly introduced by this layer (flagged "new to Layer 8") or confirmed values carried forward from an upstream layer's own reference list (flagged accordingly).

[1] Oesterhelt F, Rief M, Gaub HE (1999). Single molecule force spectroscopy by AFM indicates helical structure of poly(ethylene-glycol) in water. *New J. Phys.* 1:6. – PEG persistence length $l_p \approx 0.38$ nm; new to Layer 8, not yet confirmed against a paper physically present in `papers/layer_08/`.

[2] Marko JF, Siggia ED (1995). Stretching DNA. *Macromolecules* 28(26):8759–8770. – Worm-like-chain framework and the short-chain non-Gaussian $P(r)$ regime referenced in § II–III; new to Layer 8, not yet confirmed against a paper physically present in `papers/layer_08/`.

[3] Rant U, Arinaga K, Fujita S, Yokoyama N, Abstreiter G, Tornow M (2004). Electrical manipulation of oligonucleotides grafted to charged surfaces. *Nano Lett.* – Precedent for surface-tethered, conformationally-gated electron transfer, motivating the surface-exclusion boundary condition in § III; new to Layer 8, not yet confirmed against a paper physically present in `papers/layer_08/`.

[4] Netz RR, Joanny JF (1999). Complexation between a semiflexible polyelectrolyte and an oppositely charged sphere. *Macromolecules* 32(26):9013–9025. – Framework cited for the electrostatic bending argument in § V; not evaluated quantitatively here (Limitation 5); new to Layer 8, not yet confirmed against a paper physically present in `papers/layer_08/`.

[5] Kridel SJ, Chen E, Kotra LP, Howard EW, Mobashery S, Smith JW (2001). Substrate hydrolysis by matrix metalloproteinase-9. *J. Biol. Chem.* 276(23):20572–20578. – $K_m^{\text{sol}} = 1.28$ mM for SGKIPRTLTA, used in § VIII; confirmed via Layer 9.

[6] Smalley JF, Feldberg SW, Chidsey CED, Linford MR, Newton MD, Liu Y-P (1995). The kinetics of electron transfer through ferrocene-terminated alkanethiol monolayers on gold. *J. Phys. Chem.* 99:13141–13149. – $\beta = 0.95$ Å^{-1}, $k_0 = 6.0 \times 10^8$ s^{-1}, used throughout § IV–V; confirmed via Layer 10.

[7] Adjémian J, Anne A, Cauet G, Demaille C (2010). Cleavage-sensing peptide-enzyme multilayer gold surfaces for the parallel, selective, and sensitive assay of trypsin and plasmin activities by electrochemical methods. *Langmuir* 26:10347–10356. – Context for the Γ_p target window and functional coverage limits referenced in § VII; confirmed via Layer 7.